

Md. Habib Ullah Yeasin

Flutter Developer | Full-Stack Software Engineer

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Professional Summary

Mobile systems engineer specializing in **Flutter** and **distributed real-time infrastructure**, with **1+ years** shipping production applications to the Google Play Store and Apple App Store (**4 live on Play, 2 on the App Store**). Architects across the full stack — Flutter clients, **stateful Node.js/TypeScript services**, **Python/FastAPI microservices**, **serverless Firebase backends**, and **React operations dashboards** — unified behind shared authentication, real-time, and notification substrates. Depth in **VoIP and WebSocket infrastructure (Agora RTC, Socket.IO)**, **AI-native developer tooling (VS Code Extension API, LLM integrations)**, **payment reconciliation pipelines**, and **geospatial data engineering (PostgreSQL/PostGIS)**. Delivers systems that hold state correctly under concurrency, network failure, and third-party latency.

Technical Skills

Languages: Dart, TypeScript, JavaScript, **Python**, SQL, C++

Distributed & Real-Time Systems: Socket.IO, native WebSockets, Agora RTC (VoIP), CallKit / PushKit (iOS), FCM & APNs delivery pipelines, APScheduler background jobs, WebSocket reconnection & deduplication

Backend & API Engineering: Node.js (Node 22, ESM), Express.js, **FastAPI**, REST API design, domain-driven modular architecture, contract validation (**Pydantic**, Zod, Joi, class-validator), Swagger/OpenAPI

AI-Native Development Tooling: VS Code Extension API, LLM context engineering (Claude Code, GitHub Copilot, Gemini, OpenAI), semantic code search, dependency-graph parsing, system prompt pre-compilation

Cloud & Serverless: Firebase (Cloud Functions, Firestore, Cloud Storage, Auth, App Check), Secrets Manager, Firebase Admin SDK, AWS S3, Cloudinary, Vercel

Security, Auth & Compliance: JWT access/refresh rotation (python-jose, passlib/bcrypt), custom OTP-based 2FA, OAuth, RBAC, Firestore Security Rules, RSA/AES encryption (pointycastle), biometric/PIN enforcement, GDPR erasure, SlowAPI rate-limiting

Payments & Revenue Infrastructure: Stripe (PaymentIntents, SetupIntents, webhooks), PayPal (order create/capture, webhooks), webhook reconciliation, RevenueCat entitlement synchronization, in-app purchase

Data & Persistence: **PostgreSQL + PostGIS (GiST indexes, ST_DWithin)**, SQLAlchemy 2.0, Alembic migrations, MongoDB/Mongoose, Firestore, Redis, SQLite (sqlite), Hive, flutter_secure_storage

Mobile Engineering: Flutter (Android + iOS), native integration, iOS toolchain (CocoaPods, Swift Package Manager), deep linking, responsive UI (flutter_screenutil), i18n, PDF generation, media capture/compression

Architecture & State: Clean Architecture, Bloc/Cubit (flutter_bloc), Riverpod, Provider, GetX, dependency injection (get_it), functional error handling (dartx Either<Failure, T>), schema version tolerance

Web & Operations Tooling: React 18, TypeScript, Vite, React Router, Axios, Tailwind CSS v4, Recharts, Leaflet, service-worker push, socket-driven live dashboards, browser-native media processing

Reliability & Observability: **Pytest**, **Playwright (E2E)**, log4js structured logging, bloc.test / mocktail / flutter_test, retry/backoff strategies, forced-update version gating, Play Store permission auditing

Tools & DevOps: Git, GitHub, **GitHub Actions (CI/CD)**, Firebase CLI, Postman (**Certified**), VS Code, Android Studio

Professional Experience

Flutter App Developer | *scaleUP*

Aug 2025 – Aug 2026

- **Engineered production-grade cross-platform applications**, as measured by accelerated feature delivery and stable client releases to the Google Play Store and Apple App Store, by leveraging **Flutter** and **Clean Architecture** principles across the entire mobile lifecycle.
- **Optimized application state and data synchronization**, as measured by fluid 60-FPS rendering and seamless offline resilience, by implementing advanced **BLoC/Riverpod** state management alongside local persistence (**Hive/sqlite**) and functional error handling (dartx).
- **Integrated real-time backend services and third-party APIs**, as measured by reliable event delivery and low-latency data fetching, by systematically mapping RESTful endpoints via **Dio** and consuming **WebSocket/Socket.IO** streams for live chat and notifications.
- **Streamlined deployment and cross-team collaboration**, as measured by zero regressions in production environments,

by enforcing strict version control via **Git**, testing endpoint contracts using **Postman**, and resolving iOS/Android native configuration requirements.

Key Projects

Two One Two Messenger | *Distributed Real-Time Communications Platform*



Flutter client + stateful Node.js/TypeScript real-time services on MongoDB — owned end-to-end.

- **Architected an end-to-end distributed communications platform**, as measured by a live production release sustaining concurrent multi-device sessions, by building a **decoupled layered Node.js/TypeScript service** — strict domain separation (models/DTOs/services/controllers) from infrastructure — pairing a REST API for core business logic with a **stateful Socket.io real-time engine** that maintains group-call participant state, presence, and typing over a MongoDB persistence layer.
- **Engineered a high-availability VoIP subsystem**, as measured by sustained call signaling and delivery through background and terminated application states, by integrating **Agora RTC** with native platform call abstractions — **iOS CallKit** and **PushKit VoIP tokens** — via `flutter.callkit.incoming`.
- **Designed a zero-server-trust client-side cryptographic system**, as measured by end-to-end encryption of message payloads with no server-held decryption capability, by implementing **hybrid asymmetric/symmetric encryption (RSA + AES)** through pointycastle, governing key exchange with a **Mongoose EC (Elliptic Curve) key model**, and confining all private key material to **platform-backed secure storage** (Keychain/Keystore via `flutter.secure.storage`).
- **Enforced production API integrity and release safety**, as measured by contract-validated endpoints and version-gated clients, by constructing a **domain-driven backend pipeline** featuring strict structural middleware validation (**class-validator/class-transformer + Joi**), live auto-generated **Swagger** contracts, **daily-rotated log4js** structured logging, **forced-update version gating** protecting production APIs against breaking client-side updates, and **well-known universal-link endpoints**.
- **Developed a resilient multi-platform notification and push pipeline**, as measured by correct high-velocity multi-device delivery with payload **deduplication** and conditional call-vs-message routing across foreground, background, and terminated states, by unifying **FCM** and a dedicated native **APNs** service with CallKit hand-off.
- **Implemented an offline-first data synchronization layer**, as measured by full local message persistence via **sqlite**, layered beneath a unified state architecture of **20+ BLoC/Cubit modules under bloc.test/mocktail coverage** and a custom auto-reconnecting Socket.IO transport exposing **20+ typed event listeners** over a Dio REST fallback path.
- **Operationalized three concurrent revenue streams**, as measured by live **Stripe** payment processing, **in_app_purchase** premium entitlements, and **5 Google Mobile Ads** formats (banner, interstitial, rewarded, native, app-open), alongside **AWS S3** media infrastructure, **JWT + bcrypt + OTP** authentication, and **node-schedule cron workers** enforcing disappearing-message retention policy.
- **Cleared platform compliance and anti-exfiltration requirements**, as measured by successful Play Store review, by auditing and stripping unjustified manifest permissions and enforcing screenshot/recording protection; shipped **4-language ARB internationalization** and deep-linked invite flows.

FFP Vault — Financial Freedom Power | *Serverless Fin-Tech Architecture*



Mature, iteratively shipped financial product (v6.0.0+8) on a fully managed serverless backend.

- **Architected a fully managed serverless fin-tech platform**, eliminating custom server operational overhead entirely, as measured by a mature **v6.0.0+8** production release spanning **100+ screens** (~104 Dart files) and **over 50 nested routes**, by building a feature-partitioned Riverpod + `go_router` (`ShellRoute`) client against a pure **Firestore control plane** — Auth, Firestore, Cloud Storage, Cloud Functions, FCM, and App Check — with no REST or GraphQL tier.
- **Engineered a custom multi-factor identity layer above the managed auth provider**, extending beyond its native capabilities, as measured by **rate-limited, expiry-bound, salted SHA-256** one-time passcodes governing 2FA, registration verification, and credential recovery, by orchestrating **Cloud Functions (Node 22 ESM)** against **Secrets Manager**-held credentials with Nodemailer/SMTP as the delivery channel.
- **Eliminated a class of revenue-state corruption under distributed race conditions**, as measured by **regression guards** rejecting stale writes that would clobber active entitlements, by architecting a **bidirectional RevenueCat reconciliation pipeline** (server webhook + client sync) with propagation-delay-tolerant entitlement polling and restore-purchase recovery paths.
- **Enforced authorization as declarative policy at the data layer rather than in client logic**, as measured by ownership-scoped **Firestore Security Rules** that structurally block client-side privilege escalation against billing fields, establishing per-tenant data isolation independent of application-code correctness.
- **Delivered regulatory right-to-erasure compliance**, as measured by a single authoritative deletion path cascading across Firestore documents, Cloud Storage objects, OTP records, and the Auth identity itself, by implementing a callable Cloud Function as the **GDPR erasure boundary**.
- **Established defense-in-depth across the device trust boundary**, as measured by layered **biometric authentication**,

PIN lock, and session-timeout auto-lock over encrypted local storage, hardened at the platform layer by **App Check** attestation (Play Integrity / App Attest) validating client integrity against unauthorized backend access.

- **Automated recurring revenue and engagement operations without dedicated infrastructure**, as measured by a **15-minute reminder-notification worker** and a **daily trial-expiration worker**, by deploying scheduled serverless cron jobs against the entitlement and notification stores.
- **Modeled the domain's financial computation logic**, as measured by **loan amortization** scheduling, multi-frequency payment normalization, and housing/insurance cost tracking, exposed alongside a **subscription-gated document Vault** with full file-lifecycle CRUD (upload/download/share) over Hive-backed local persistence with code-generated adapters, Google Sign-In, and Sign in with Apple.
- **Unblocked a stalled iOS release pipeline**, as measured by resolution of a **Swift Package Manager duplicate-symbol collision**, by isolating the dependency conflict and codifying a documented Podfile pin as the permanent remediation.

Hesteka | *Three-Tier Platform: Mobile, API, and Operations Dashboard*



Sole engineer across all three tiers, unified by one auth, real-time, and notification substrate.

- **Owned the complete three-tier platform architecture end-to-end**, as measured by a Flutter client, a **Node.js/TypeScript/Express/MongoDB** API, and a **React 18** operations console all operating on one shared identity, real-time, and notification substrate — every tier delivered by a single engineer.
- **Engineered an automated payment reconciliation pipeline spanning two independent gateways**, as measured by verified webhook settlement for **Stripe** (PaymentIntents, SetupIntents, saved payment methods) and **PayPal** (order create/capture), by implementing robust exception handling, structural validation, and dedicated harnesses exercising the reconciliation path against the live API.
- **Institutionalized backend structural consistency and delivery velocity**, as measured by **20 uniformly layered feature modules** (controller/service/route/model/validation), by authoring a custom **code-generation tool** (`makeModule.ts`) that scaffolds compliant modules by default and mandating **Zod contract validation at every endpoint boundary**.
- **Secured a federated multi-tenant identity system**, as measured by **JWT access/refresh token rotation**, Google OAuth, Sign in with Apple, and OTP email verification across **three role classes** (admin/user/partner), by layering bcrypt hashing, account-status gating, **Redis-backed rate limiting**, and CORS allow-listing as sequential defensive controls.
- **Built an authenticated real-time messaging substrate with native trust-and-safety primitives**, as measured by community and private 1:1 channels carrying report/like moderation actions, by enforcing **JWT verification in custom Socket.IO middleware** at connection establishment rather than per-message.
- **Operationalized a points and rewards ledger economy**, as measured by a **cron-driven balance-reset cycle** and photo-based donation-proof validation gating credit issuance, by binding verified proof submissions directly to ledger mutations.
- **Implemented spatial indexing for proximity-driven engagement**, as measured by **geo-fenced “local missions”** resolved against user position, by applying **geohash-based radius querying** in place of unindexed distance scans.
- **Synchronized an external commerce catalog into the platform**, as measured by product and collection data mirrored consistently into both mobile and administrative surfaces, by integrating the **Shopify Admin API**; media offloaded to Cloudinary and transactional mail routed through Nodemailer/SMTP.
- **Delivered the internal operations console**, as measured by **18 CRUD administration surfaces**, by building React 18 + Vite + Tailwind v4 atop reusable DataTable/CRUDModal/FilterBar primitives with **Recharts** analytics, a live **Leaflet** map of geo-tagged field reports, socket-driven state updates, and service-worker push parity with mobile.
- **Unified push-delivery semantics across three heterogeneous clients** (Node backend, React admin service worker, Flutter mobile) through a single **FCM** integration; mobile built on code-generated Riverpod providers, **go.router route guards**, Clean Architecture (`core/ + features/{data,domain,presentation}`), `freezed + json.serializable`, Dio, and `flutter_secure_storage`, with FR/EN localization implemented independently in **two distinct paradigms** (React context and Riverpod notifier).

DocMobi | *Telehealth Consultation Infrastructure*



Role-partitioned (patient/doctor) clinical platform on a fully custom REST backend.

- **Engineered a fault-tolerant API client layer**, as measured by automatic JWT refresh, centralized 401 session invalidation, **retry/backoff**, and multipart transfer across **7 domain service modules** (auth, appointments, chat, doctors, users, dependents, uploads), by hand-building a unified `ApiClient` abstraction rather than dispersing raw HTTP calls.
- **Architected a real-time clinical consultation stack**, as measured by live audio/video sessions with channel management and full call controls (mute, camera switch, speaker toggle), by layering the **Agora RTC** media plane over a **Socket.IO** signaling, chat, and presence control plane.
- **Delivered OS-level call interception**, as measured by call screens rendering over the device lockscreen, by configuring custom high-importance Android notification channels with **full-screen intents**.
- **Reduced third-party mapping dependency and cost exposure**, as measured by nearby-provider discovery ranked via

Haversine distance, by implementing **OpenStreetMap** through `flutter_map` with a custom cached tile provider, custom marker factory, and geocoding — bypassing the default Google Maps SDK.

- **Built a clinical document pipeline**, as measured by end-to-end handling from capture to archive across Android and iOS, by chaining image capture/compression with **PDF generation, printing, and in-app viewing** for medical records.
- **Routed push notifications into arbitrary application states**, as measured by deterministic deep-link navigation from foreground, background, and terminated states, by dispatching through a **global navigator key** outside the widget tree.
- **Served two role-partitioned product experiences from a single codebase**, as measured by fully separate patient and doctor interfaces, by implementing role-based routing and theming over **6 domain providers** and singleton cross-cutting services (sockets, calls, location, notifications).
- **Modeled multi-actor healthcare workflows**, as measured by appointment scheduling, medical record management, dependent/family-member booking, provider reviews, and emergency contacts, with **Hive** caching enabling offline appointment access.

SnapShip | *Multi-Vendor Logistics Aggregation & AI Order-Intake Engine*



Merchant dispatch hub normalizing three national courier networks behind one interface.

- **Normalized a fragmented logistics vendor landscape**, as measured by **3 national courier APIs** (Pathao, RedX, Steadfast) with incompatible credential schemas — OAuth-style client/secret versus static API keys — unified behind a single domain interface with **live rate comparison** before dispatch.
- **Automated unstructured order intake**, as measured by a **confidence-scoring system** (overall score, address-match score, match method, phone validity) surfacing human-readable warnings before dispatch, by building the “SnapEngine” AI parsing pipeline that converts free-text **mixed Bangla/English** merchant messages into structured shipping records (name, phone, address, COD amount, weight).
- **Eliminated unchecked exceptions across async boundaries**, as measured by explicit `Failure` types propagated end-to-end from repository to UI, by adopting **dartx** `Either<Failure, T>` functional error handling in place of throw/catch.
- **De-risked vendor credential onboarding**, as measured by **live “Test Connection” validation** against real courier auth endpoints prior to persistence, by building an encrypted credential vault on `flutter_secure_storage` with per-courier webhook configuration for delivery-status callbacks.
- **Future-proofed API consumption against upstream schema drift**, as measured by backwards-compatible parsing of both **v1 flat and v2 nested response shapes**, by enforcing a `Model→Entity` mapping boundary at the repository layer.
- **Enforced testable, decoupled composition**, as measured by a **single `get_it` composition root** registering singletons and factories across **5 Clean Architecture feature modules** (auth, dashboard, profile, `ship_hub`, `snap_engine`) with `BLoC` for complex async flows and narrow `Cubits` for scoped concerns.
- **Centralized transport-layer concerns**, as measured by a single `Dio` client handling auth-token injection, request/response logging, and 401 session invalidation, with **flutter_dotenv** externalizing secrets and base URLs out of source control; shipment tracking delivered via a dedicated tracking `BLoC/repository`.
- **Codified a reusable design system**, as measured by dedicated `AppTheme/AppColors/AppTypography` layers, dark-mode-first components, `CustomPainter`-rendered borders, glassmorphism cards, and shimmer loading states.

Language Learning App | *EdTech Platform*



Interactive educational application for vocabulary and grammar acquisition.

- **Implemented LMS-grade learning mechanics**, as measured by interactive quizzes, progress tracking, and daily streak retention loops, by modeling assessment and progression logic comparable to modern learning management platforms.
- **Maintained consistent learner state across sessions**, as measured by persisted user sessions, quiz scores, and streak counters, by centralizing state in **Riverpod** providers.
- **Delivered an accessible learning interface**, as measured by a responsive, student-oriented UI with audio playback for pronunciation practice.

LetzGo | *Ride-Sharing Marketplace Platform (Full-Stack)*

[App](#) • [Backend](#) • [Admin](#)

Three-tier ride-pooling platform: Flutter mobile app, FastAPI backend, React admin dashboard.

- **Architected a full-stack ride-pooling marketplace**, as measured by a complete three-tier deployment, by developing a **Flutter** mobile client, a **FastAPI/Python** REST and WebSocket backend, and a **React 18 + TypeScript** administration dashboard.
- **Engineered real-time nearby-ride discovery**, as measured by low-latency spatial queries against active drivers, by designing a **PostgreSQL + PostGIS** schema (leveraging `ST_DWithin` and `GiST` indexing) mapped via **SQLAlchemy 2.0** ORM and **Alembic** migrations.
- **Orchestrated live ride-state synchronization and in-app chat**, as measured by zero-polling event delivery, by implementing native **WebSockets**, background **APScheduler** workers for ping expiry and analytics rollups, and **SlowAPI** rate-limiting to prevent endpoint abuse.
- **Secured platform access and notifications**, as measured by role-based dispatch routing, by integrating **JWT** authenti-

cation (python-jose, passlib/bcrypt) and the Firebase Admin SDK for phone verification and FCM push delivery.

- **Delivered an operational moderation console**, as measured by fully tested CRUD administration flows backed by **Pytest** and **Playwright** end-to-end (E2E) testing coverage, by building a frontend web suite utilizing Vite, React Router, Axios, and Recharts.

PromptIR | AI-Native Developer Tooling (VS Code Extension)



Context-aware prompt pre-compiler for AI coding assistants — published and live.

- **Engineered an AI-native developer tooling extension**, as measured by live publication on the VS Code Marketplace, by building a context-aware prompt compiler that aggregates the active file, diagnostic signals, and semantic code searches into structured, agent-ready prompts for **Claude Code**, **Gemini**, and **GitHub Copilot**.
- **Accelerated developer refactoring workflows**, as measured by standardized LLM interactions, by shipping **11 specialized prompt presets** (debugging, security/performance review, UI/UX improvement, implementation planning) managed via an integrated sidebar UI.
- **Established flexible LLM provider compatibility**, as measured by dynamic model switching, by integrating **GitHub Copilot's native Language Model API** and a configurable **OpenAI** provider, enhanced with dependency-graph-aware context mapping for larger repositories.
- **Ensured local data privacy and security**, as measured by zero unauthorized telemetry, by confining context processing locally to the editor and securing API keys via VS Code's Secret Storage; successfully published **open source under the MIT license**.

ToolsDigger | Local-First Developer Utility Platform



Browser-native file and prompt tooling — no signup, no tracking, no server uploads.

- **Architected a local-first web utility platform**, as measured by zero server-side user data processing, by developing client-side web tools for prompt compilation, file conversion, and code-showcase workflows with no account requirement.
- **Engineered a browser-native media processing pipeline**, as measured by direct frame encoding, by building “**Zip to Video**”, a tool that converts a ZIP archive of image frames into a shareable, self-extracting MP4 entirely on the client.
- **Expanded the AI-prompting ecosystem**, as measured by standalone composition capabilities, by shipping a web counterpart to PromptIR with specialized compression and packaging tools tailored specifically for prompt engineers.
- **Delivered a comprehensive open-source tooling catalog**, as measured by **199 indexed no-signup web utilities** across 10 categories, extending platform utility and discovery beyond its natively built tools.

Education

University of Frontier Technology, Bangladesh

Jan 2020 – Jan 2025

B.Sc. in Educational Technology and Engineering (EdTEE)

CGPA: 3.28/4.00

A multidisciplinary program integrating software development, instructional design, networking, and engineering principles.

Relevant Coursework: Mobile Application Development, Data Structures & Algorithms, OOP, Database Management Systems, Networking, Software Engineering, Database Systems, Cloud Computing, Testing & QA.

Certifications

- **SQL (Basics) Certified (HackerRank)** – Validates strong relational database knowledge.
- **Postman API Fundamentals Student Expert – Postman, 2025**
- **AWS Academy Graduate: Introduction to Cloud (2025)**
- **CS Fundamentals With Phitron (2025)** – Focus on C++ and Data Structures.
- **Soft Skill: 21st Century Core Employability Skills Development Program – Wadhwani Foundation, 2023**
Achieved 90% score in the “Competencies of Job Ready Modules” under Wadhwani Opportunity’s global workforce readiness initiative.